

Sage X

Architecture

There is an broadcaster which encrypts file So the broadcast have a database in which a user ID concealed. So when a user sign in, the user has to do something called at key integration. So the user will uh sign in and uh the user will get create a post quantum cryptographic private public key. So private key will can will stay in his computer and the public key will be go to a CA or certificate certificate authority. Then the certificate authority have its own key and the certificate authority will Shine the public key and send it to the broadcaster. And then the broadcaster will store it. Okay. So when the file required. to be encrypted at first a random number is generated then the random number will work as a a e s 256 data encryption key or b e K. So the DK will encrypt the file and then the DK will DK will encrypt it with post quantum cryptos. graphic or M L K E G. So this is term with the public with the use of signed public key for each of the user. So that's how the encryption All right.

After encryption, the recipient will get its file. So when the file is decrypted with its private key, it will generate bytes then we will do multi-level watermarking. Okay, the first level is modern directly metadata. Second level is byte level encryption. And third level is something called as stenographic stenographic encryption watermarking. Third is geometric line. Fourth level is Osine Transform Transformation. DWT or C CWT? O sine wave transformation or D. W T dispute web transformation in this format the file will be watermarked.

After the watermark generation, the watermark user ID session ID and the timestamp will be converted into a hash. It is generic non non-cryptographic hash. Then the cryptographic hash will be signed with

public key. Then the cryptography will be stored in a distributed ledger system or a blockchain system. That's how it cannot be tempered later. That's it.